PRODUCT DATA SHEET

METAL WORKING FLUID – TECHNOFLUID CUTTING OIL

Commercial / Pack Name: CUTTING OIL (WATER SOLUBLE)

Product Description

- · TECHNOFLUID Cutting Oil is a high-performance metal working fluid formulated to provide effective lubrication, cooling, and surface protection during machining operations.
- · Manufactured using selected high-quality base oils and advanced additive technology to ensure superior cutting performance.
- · Designed to reduce friction and heat generation at the cutting zone, thereby improving tool life and machining accuracy.
- · Available in soluble and neat (straight) cutting oil variants to suit a wide range of machining applications.
- · Ensures clean operation with stable performance over extended service life.

Key Performance Benefits

- · Excellent lubricity reduces tool wear and improves surface finish of machined components.
- · Effective heat removal minimizes thermal distortion during machining.
- · Superior anti-wear and extreme pressure properties support heavy-duty cutting operations.
- · Good emulsion stability (for water-soluble grades) ensures consistent performance.
- · Provides effective rust and corrosion protection to machined parts and machine components.
- · Operator-friendly formulation with low odor and minimal misting.
- · Reduces tool breakage, rejects, and overall machining costs.

Applications

Recommended for a wide range of metal machining operations including:

- Turning, milling, drilling, and boring
- Grinding operations
- Tapping and threading

Suitable for machining of:

- Mild steel and carbon steel
- Alloy steels
- Cast iron
- Selected non-ferrous metals (as applicable)

Widely used in automotive, engineering, fabrication, and metal processing industries.

Typical Uses

- · General-purpose machining operations requiring reliable lubrication and cooling.
- · Medium to heavy-duty cutting where improved tool life and surface finish are critical.
- · CNC and conventional machine tools.

Operating Characteristics

- · Maintains stable viscosity and lubricating film under operating conditions.
- · Compatible with commonly used machine tool materials and seals.
- · Can be applied by flood, spray, or central lubrication systems (as applicable).

Typical Characteristics

Characteristics	Test Method	Typical Figures
Appearance (Concentrate)	Visual	Clear / Light Amber
Appearance (5% Emulsion)	Visual	Milky White
Base Fluid Type	—	Mineral Oil Based (Emulsifiable)
Emulsion Stability (5% in water)	IP / In-house	Stable
Density @ 15°C, g/ml	ASTM D4052	0.875
Kinematic Viscosity @ 40°C, cSt	ASTM D445	40 – 45
pH of 5% Emulsion	ASTM D1287	8.5 – 9.0
Refractometer Factor (5% Emulsion)	—	4 – 5
Flash Point (COC), °C	ASTM D92	≥ 180
Pour Point, °C	ASTM D97	-6
Copper Corrosion (3 hrs @ 100°C)	ASTM D130	Pass
Rust Prevention (Chip Test)	ASTM D4627	Pass

Foam Tendency	ASTM D892	Low
Hard Water Stability	In-house	Good
Biostability	In-house	Good
Operating Dilution Ratio	—	3% – 10%